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Japan FX

What might happen with the JPY were a Nankai Trough megaquake to occur?

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CITI'S TAKE

The Japan Meteorological Agency (JMA) issued a warning for the Nankai Trough for the first time ever after the Hyuganada Sea earthquake last week. If a Nankai Trough megaquake were to happen, we think the JPY would appreciate significantly at first backed by Japanese investor repatriation and reinsurance payments. This initial JPY appreciation could be stronger and longer than after the 2011 earthquake and tsunami. After an initial appreciation we think the JPY would shift to deprecation via a deterioration in the balance of payments (BoP) and JGB downgrades, and this could continue until the BoJ bolsters monetary tightening to counter JPY deprecation and inflation.

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Nankai Trough megaquake

A Nankai Trough megaquake has occurred on Japan's Pacific Ocean side with the epicenter at the plate boundary from Suruga Bay to the Hyuganada Sea with an interval of around 100 to 150 years. The government says the likelihood of an occurrence over the next 30 years is 70%-80%.

According to a 2019 estimate by the government, the economic damage from such a quake could reach ¥213trn (approximately 35% of Japan's nominal GDP) in the worst case (the base case is ¥130trn), and the death toll could reach 230,000. The damage was said to be ¥16trn to ¥25trn after the 2011 earthquake and tsunami and the dead and missing totaled some 22,000 (Figure 1).

Figure 1. Major earthquakes in Japan

Month/Year		Jan-95	Oct-04	Mar-11	Apr-16	Jan-24
Disaster		Great Hanshin-Awaji Earthquake	Chuetsu Earthquake	Geat East Japan Earthquake	Kumamoto Earthquake	Noto Peninsula Earthquake
Magnitude		M7.3	M6.8	M9.0	M7.3	M7.6
Highest intensity scale		7	7	7	7	7
Personal suffering	Dead/Missing	6,437	68	22,288	273	302*
	Injured	43,792	4,805	6,233	2,809	1327*
Residential damage	Full destroyed	104,906	3,175	121,996	8,667	6227*
	Half destroyed	144,274	13,810	282,941	34,719	20,589*
Damage estimate		JPY 9.6 - 9.9tn	JPY 3tn	JPY 16 - 25tn	JPY 2.4 - 4.6tn	JPY 1.1 - 2.6tn
Supplementary budget (total size)		JPY 7.0tn	JPY 4.7tn	JPY 16.9tn	JPY 3.5tn	NA

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Note: As of July 30, 2024.

Source: Cabinet Office, MoF, Fire & Disaster Management Agency, Japan Meteorological Agency, Citi Research.

Disasters and USDJPY

Figure 2 plots the USDJPY over the long term and shows its relationship with the major earthquakes mentioned above. We note that the USDJPY and indeed any market price reflects complexity in the real world and should be seen as the result of multiple simultaneous functions. However, in the past the USDJPY has tended to ultimately rise even if it dips directly after big earthquakes. Figure 3 shows Japanese equities (TOPIX), which tend to rise after an earthquake.

Figure 2. USD/JPY and major earthquakes

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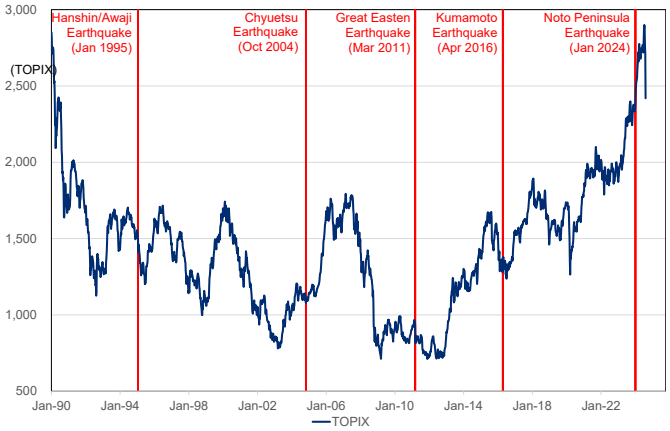
Source: Bloomberg, Citi Research.

Figure 3. Japanese equity (TOPIX) and major earthquakes



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Reconstruction demand and the trade balance

While only a single reason, one factor behind this trend in the past is likely reconstruction demand. Earthquakes tend to temporarily disrupt economic activity, but once reconstruction starts domestic demand tend to increase, partly financed by the government fiscal expenditure.

Figure 4 plots Japan's trade balance, which has deteriorated due to growth in imports after each of the earthquakes noted above. After a major natural disaster such as the 2011 earthquake and tsunami, many industrial plants are forced to shut down and bottlenecks in the supply chain cause declines in exports, so the trade balance deteriorates further.

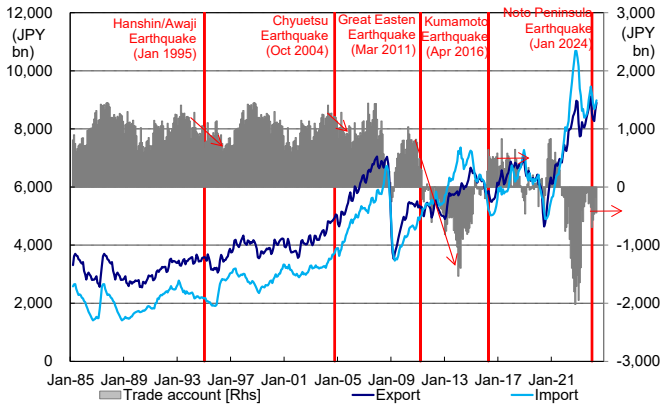
Experience from the 2011 earthquake and tsunami

After the 2011 earthquake and tsunami the JPY appreciated initially. At that time, the USDJPY was close to ¥80/$ after the decline from 2007, and a strong yen outlook had been dominant in the market even before the earthquake (Figure 5). With a natural disaster occurring under these circumstances, the view that Japanese investors would repatriate money and promote JPY strength became the consensus quickly.

Given annual mortality of about 1.1mn in Japan, however, it seemed unlikely that life insurance companies and other domestic investors would sell overseas assets at a large scale in light of the expected death toll from the 2011 disaster. Also, payments at pension funds were not expected to increase after such a natural disaster, while banks were ready to borrow JPY from the BoJ for unexpected payments to depositors.

In short, the view that the repatriation would strengthen the Japanese currency after the disaster seemed to be somewhat illusory. A more probable story was thought to be JPY depreciation due to deterioration in the balance of payments (BoP), as noted above.

Figure 4. Japan: Trade balance (three-month average)



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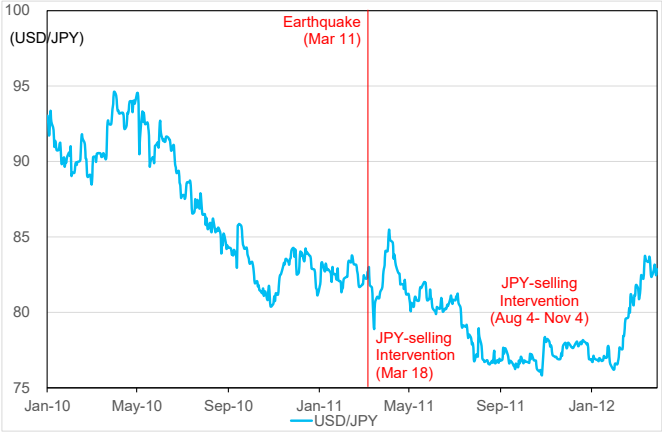
Source: MoF, Bloomberg, Citi Research.

Figure 5. USD/JPY (January 2010 – March 2012)



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Difference with the 2011 earthquake and tsunami

Despite a sharp drop in the USDJPY over the past month, the structural JPY weakness story is still persistent in Japan. Were a Nankai Trough megaquake to occur under these circumstances, a weak JPY outlook could become more dominant with the memory of prolonged JPY deprecation that has continued until today after the 2011 earthquake and tsunami.

Given the enormous potential damage, however, there is a possibility that large-scale repatriation by life and casualty insurance companies would actually occur. Payments of reinsurance would also be expected to fuel JPY-buying demand. Therefore, we suspect the initial reaction this time would be a strong JPY bounce over the time horizon of a few weeks. In the case of major disruptions in overseas markets JPY strength could continue for several months.

Even so, eventually strong downward pressure on the Japanese currency would likely emerge due to deterioration in the BoP and increasing concern for JGB rating downgrades. Also, geopolitical tensions (including the Taiwan strait risk) could escalate at the time. It seems reasonable to assume that such JPY weakness would continue at least one or two years until the BoJ bolsters monetary tightening to counter inflation.

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